Introduction to Computational Intelligence

SCJ3553 / SCK2263

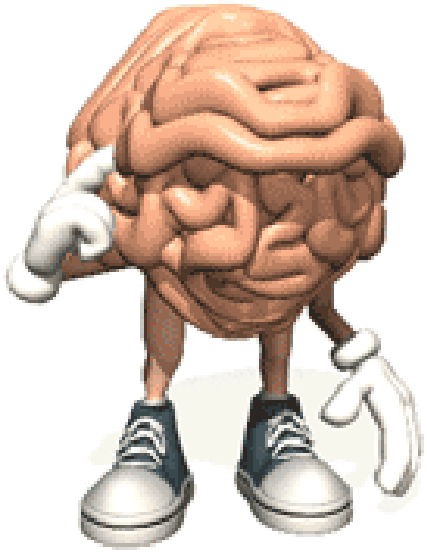
Artificial Intelligence

Outline

- What is AI?
- Now, come Computational Intelligence (CI)
- Scope of CI
- Importance of CI
- CI Techniques
  - Artificial Neural Network
  - Genetic Algorithm
  - Fuzzy Logic

What is AI?

- Ability to interact with the real world
- Reasoning and Planning
- Learning and Adaptation



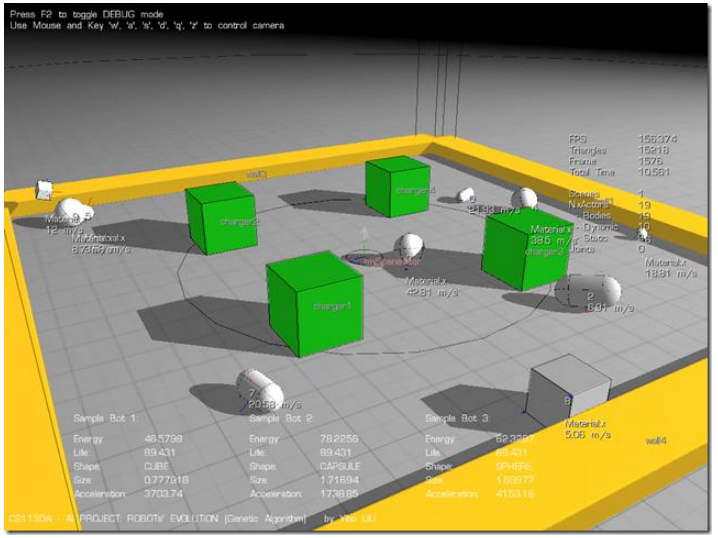
Now, come Computational Intelligence (CI)

- A methodology involving computing that exhibits an ability to learn and/or to deal with new situations, such that the system is perceived to possess one or more attributes of reason, such as generalization, discovery, association and abstraction (Russel et al, 1998).
- CI is the field of computing that draws from the successes of natural systems to develop new ways of solving computational problems in the real world.

Scope of CI

- practical adaptation concepts, paradigms, algorithms and implementations that enable or facilitate appropriate actions (intelligent behavior) in complex and changing environments.



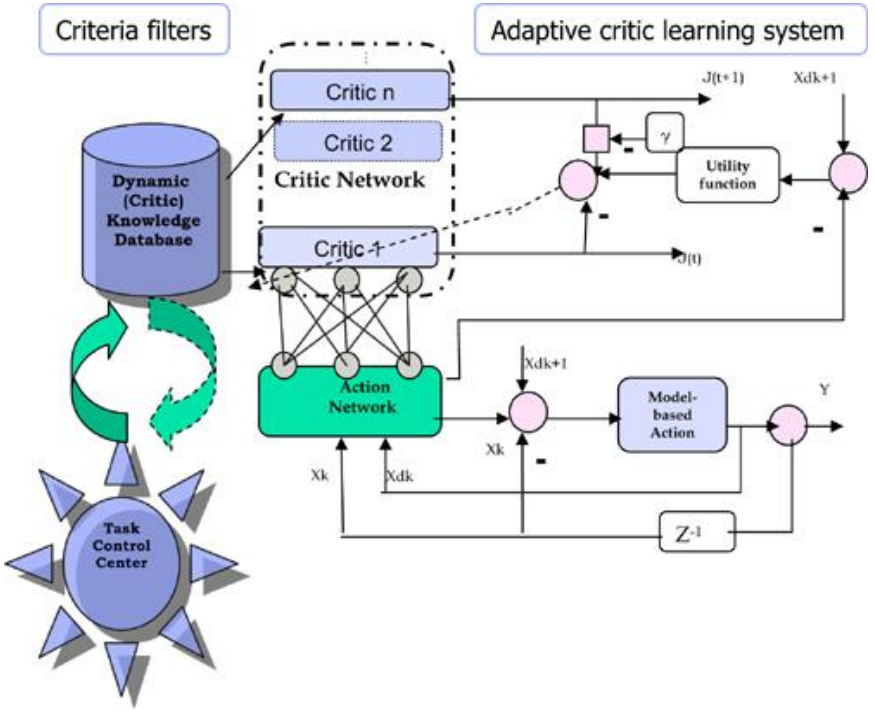


Scope of CI

- Fuzzy, imprecise or imperfect data
- No available mathematical algorithm
- Optimal solution unknown
- Rapid prototyping required
- Only domain experts available
- Robust system required
- Learning capability

More recently:

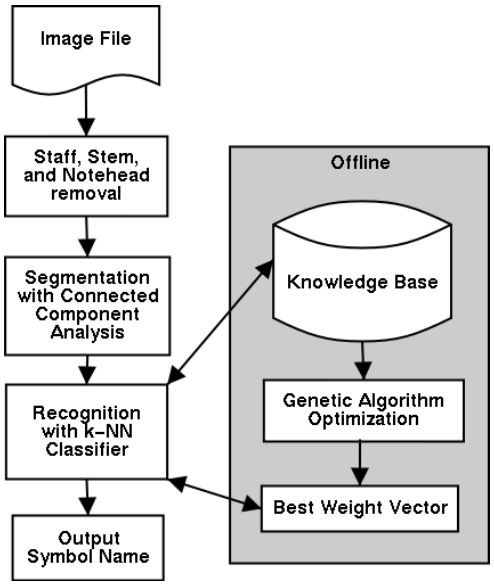
- DNA Computing
- Quantum Computing

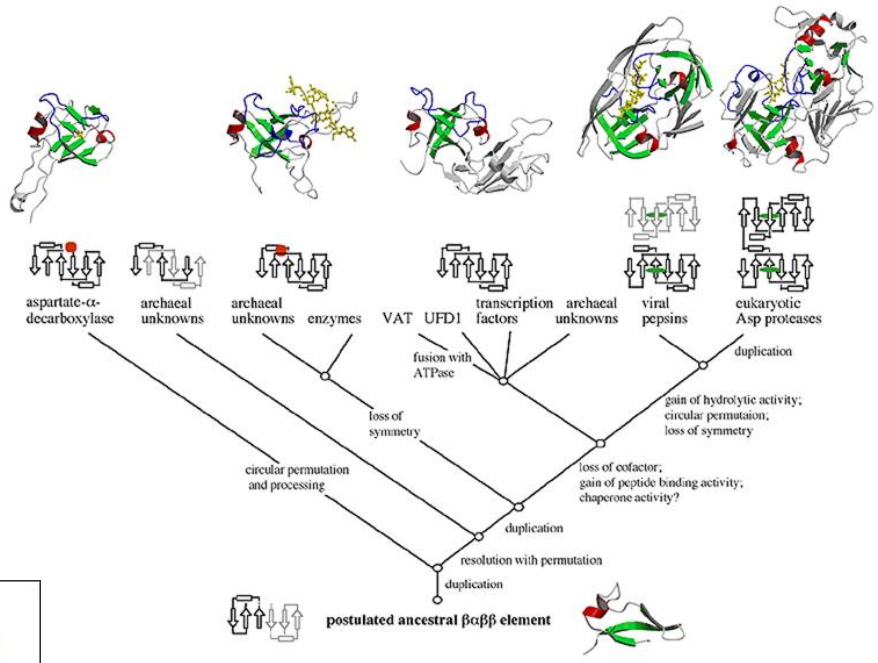


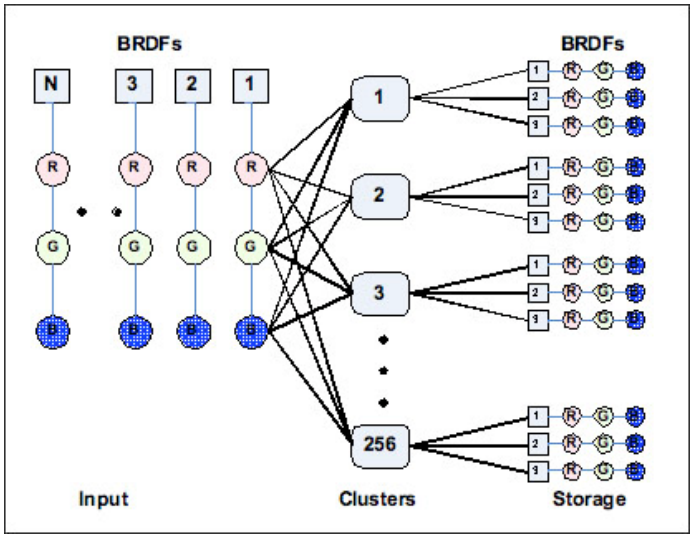
Importance of CI

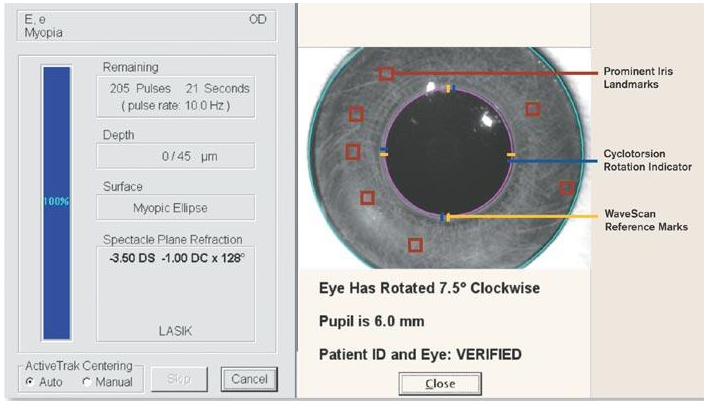
- Solve the problems in realistic situation (not ideal) such as fuzzy, imperfect or incomplete, only domain experts available.
- Complementary approaches with mathematical based algorithm solutions
- Encourage new technology development especially in Quantum and DNA computing.

Importance of CI





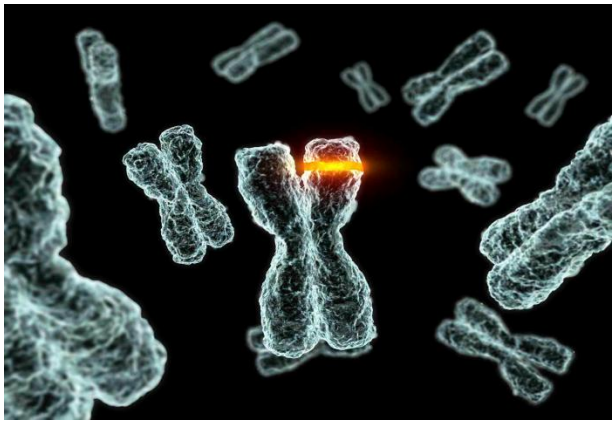




CI Techniques

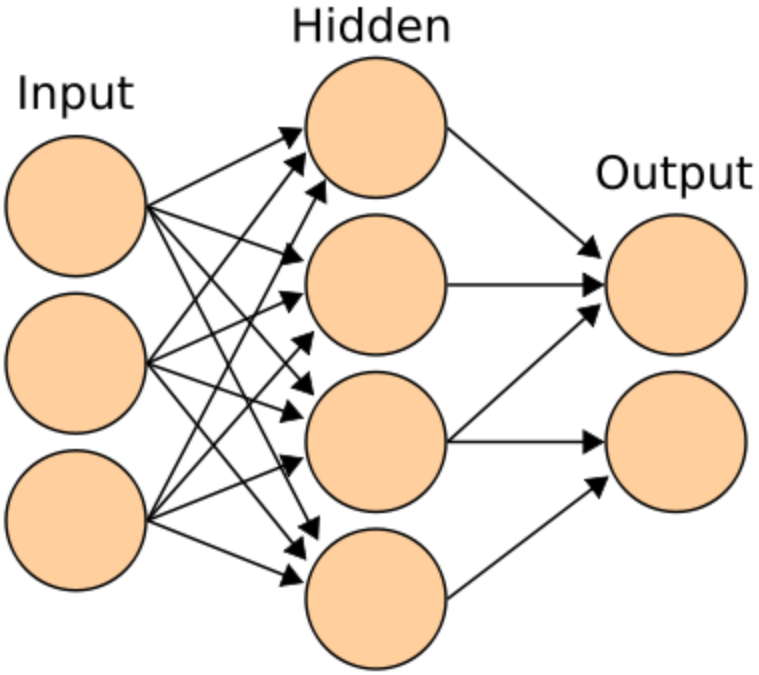
- Artificial Neural Network
- Genetic Algorithm
- Fuzzy Logic



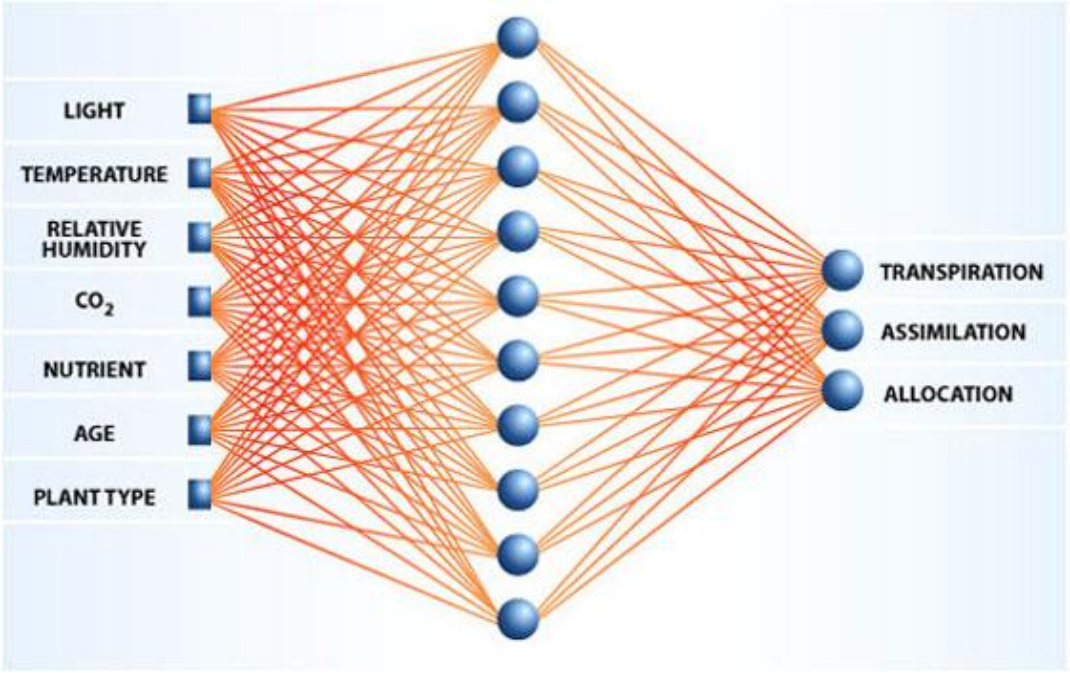




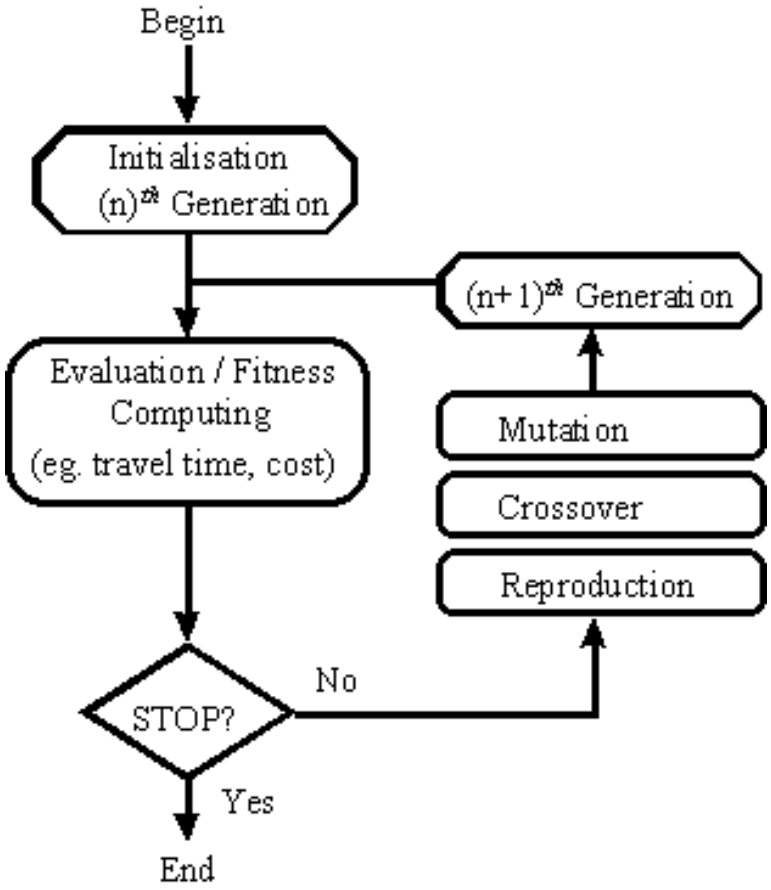
Artificial Neural Network







Genetic Algorithm



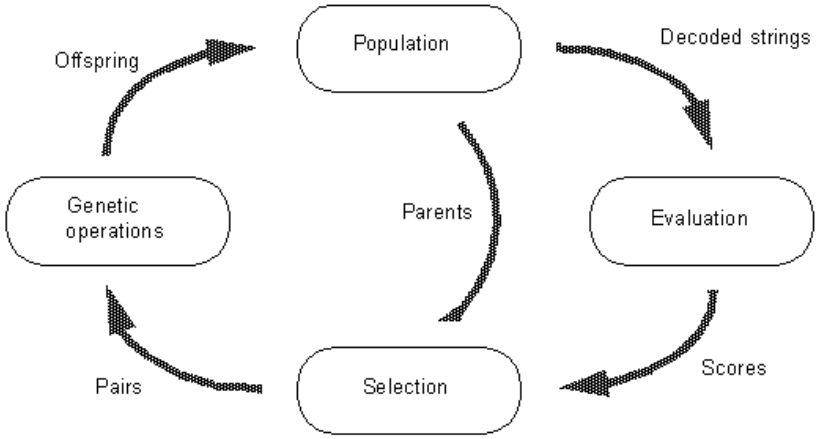
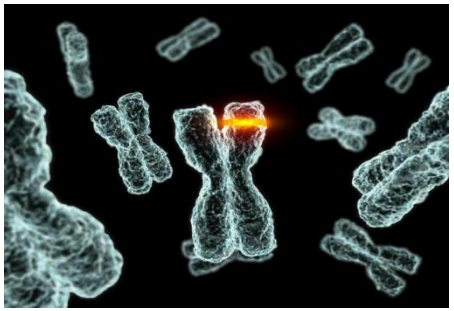
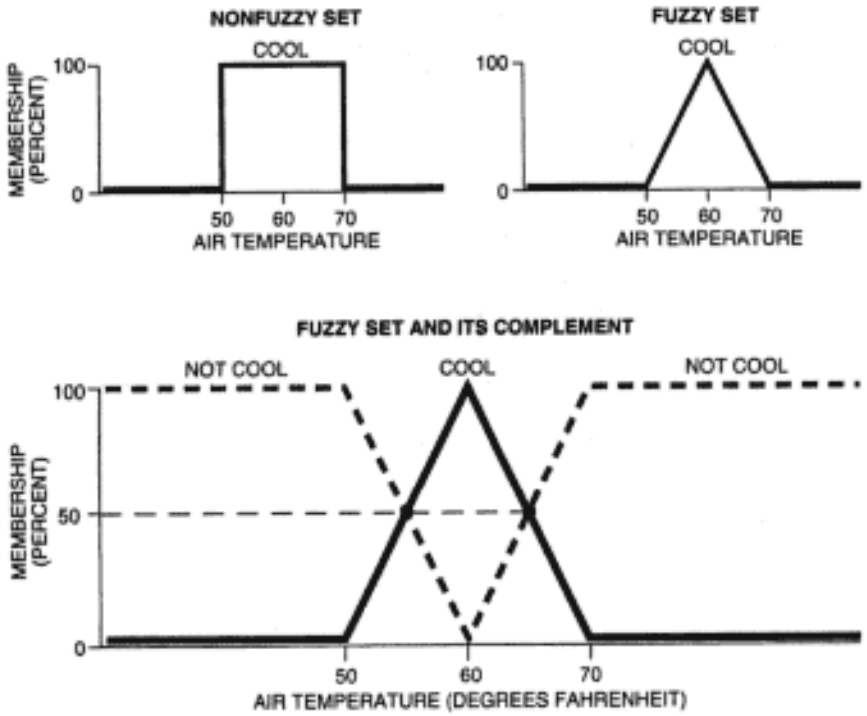
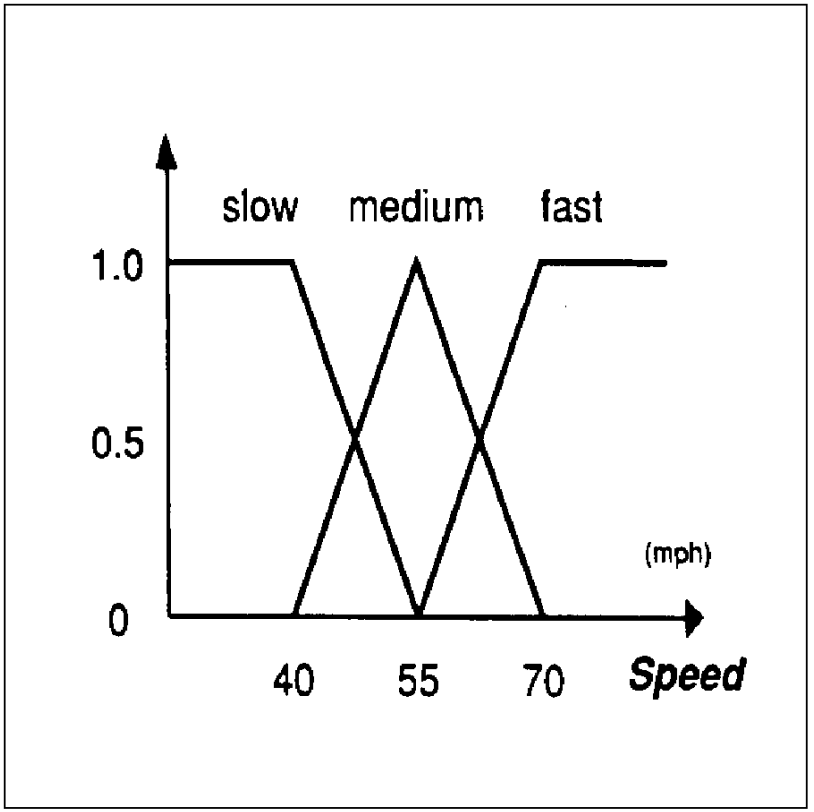


Figure 5.2: The "reproduction" cycle.

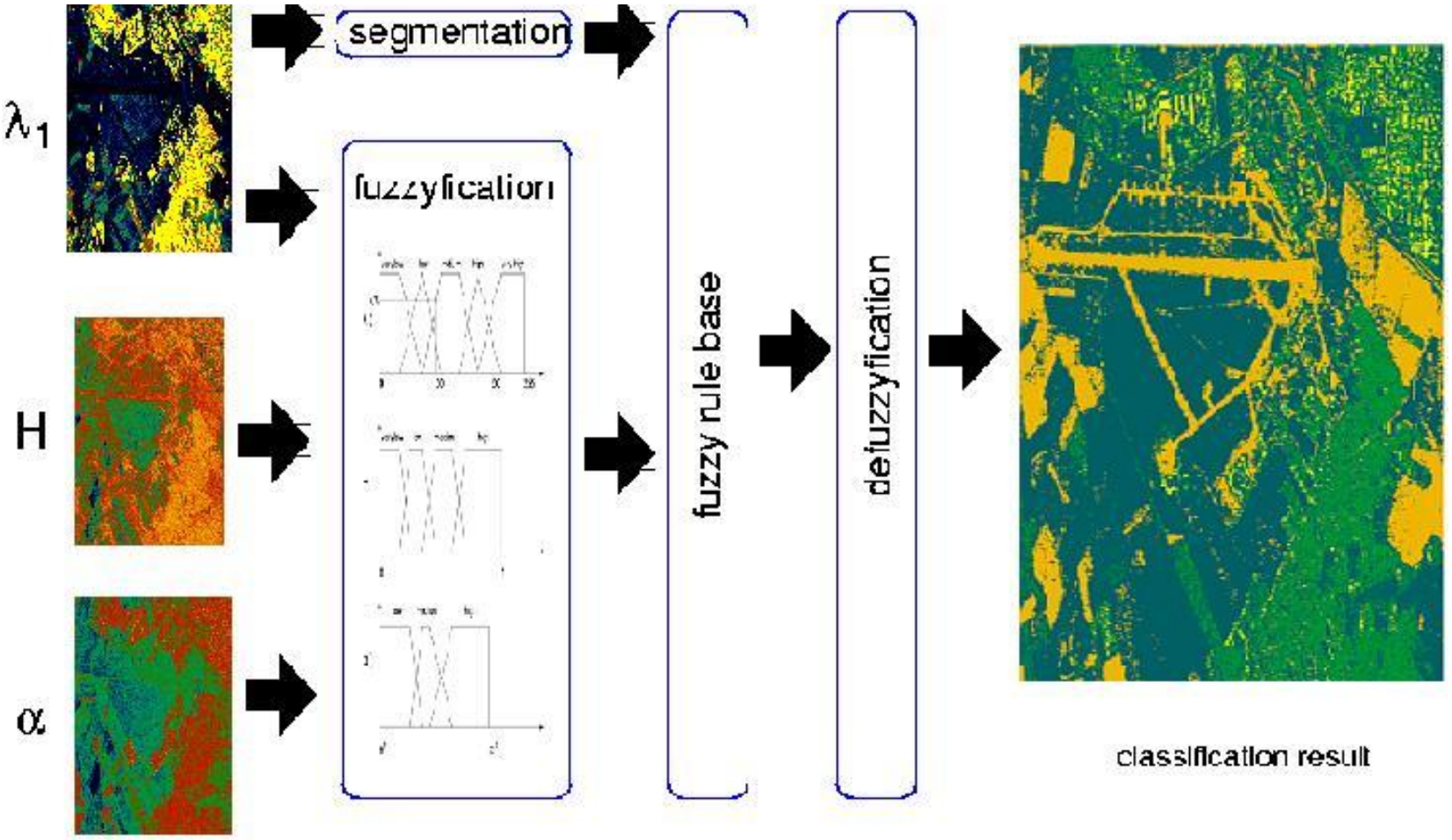


Fuzzy Logic


NONFUZZY SET
COOL
MEMBERSHIP (PERCENT)
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET
COOL
100
0
50
60
70
AIR TEMPERATURE
FUZZY SET AND ITS COMPLEMENT
NOT COOL
COOL
NOT COOL
MEMBERSHIP (PERCENT)
100
50
0
50
60
70
AIR TEMPERATURE (DEGREES FAHRENHEIT)


slow
medium
fast
1.0
0.5
0
(mph)
40
55
70
Speed

Fuzzy Logic


λ_1
H
α
segmentation
fuzzyfication
fuzzy rule base
defuzzyfication
classification result

Q & A


Panasonic
12:30
Start
Menu Select
Timer
Hour/min
Cooking Timer